

Satellite Structural Inspection Interferometric Synthetic Aperture Radar (InSAR) Based

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METHODOLOGY

1. Satellite SAR System with Resolution and Revisit Time

To select a suitable SAR system, technical specifications such as spatial resolution, frequency band, polarization, and revisit time were considered. Sentinel-1, operating at C-band (5.405 GHz), was selected due to:

Parameter	Sentinel-1 C-SAR (IW Mode)
Frequency Band	C-band (~5.405 GHz)
Revisit Time	12 days (single satellite)
Spatial Resolution	5 m (range) × 20 m (azimuth)
Coverage Mode	Interferometric Wide Swath (IW)

2. Retrieving Multi-Temporal SAR Imagery Over the Specified Period

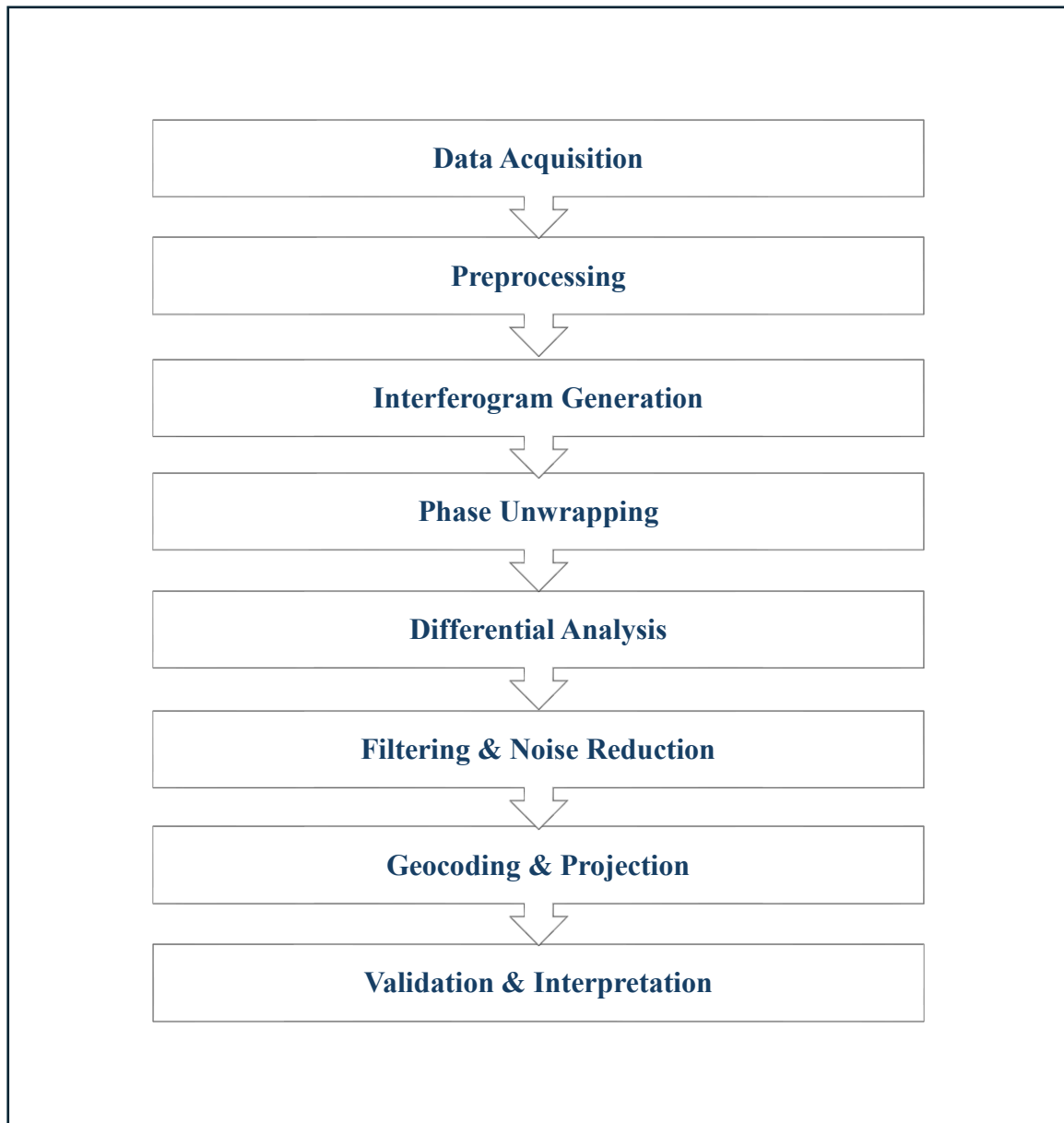
Multi-temporal **Sentinel-1 SLC IW mode** data were downloaded based on:

- **Area of Interest (AOI):** A polygon covering the bridge and surrounding terrain.
- **Date range:** Multiple acquisitions were collected spanning several months to capture temporal variations.
- **Image acquisition conditions:** All images were selected with the same relative orbit, pass direction (ascending or descending), and polarization (typically VV/VH) to ensure consistency for interferometric analysis.

S1A_IW_SLC__1SDV_20250111T130240_20250111T130310_057397_0710AB_813D
S1A_IW_SLC__1SDV_20250123T130239_20250123T130309_057572_07179E_5431
S1A_IW_SLC__1SDV_20250204T130238_20250204T130308_057747_071E86_3FE6
S1A_IW_SLC__1SDV_20250216T130238_20250216T130308_057922_072592_C312
S1A_IW_SLC__1SDV_20250228T130238_20250228T130307_058097_072CB4_7A14
S1A_IW_SLC__1SDV_20250312T130238_20250312T130308_058272_0733CA_217F
S1A_IW_SLC__1SDV_20250324T130238_20250324T130308_058447_073AB5_B445
S1A_IW_SLC__1SDV_20250405T130238_20250405T130308_058622_0741CE_2B52
S1A_IW_SLC__1SDV_20250417T130238_20250417T130308_058797_0748F4_3FFD
S1A_IW_SLC__1SDV_20250429T130239_20250429T130309_058972_07501E_08D8
S1A_IW_SLC__1SDV_20250511T130238_20250511T130308_059147_0756BA_DE13
S1A_IW_SLC__1SDV_20250523T130238_20250523T130308_059322_075CCB_14B2
S1A_IW_SLC__1SDV_20250604T130237_20250604T130307_059497_0762D9_3AB5
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S1A_IW_SLC__1SDV_20250628T130236_20250628T130306_059847_076EE3_6BFD

3. Differential SAR Interferometry (D-InSAR) for Displacement Measurement

D-InSAR analysis was performed to measure millimetre-scale deformation over time:



VISUALIZATIONS

1. Defining specific bridge location & parameters for satellite SAR based inspection



Project & Bridge Details

State – Rajasthan
Bridge No. – 899/1
Bridge Chainage – 916+750

Project - Chittorgarh to Kota

Zone – RJ 06

Structure Type – MNB

Sar Parameters

Sensor – Sentinel-1 (C-Band)
Beam Mode – IW (Interferometric Wide Swath)
Path – 34
Frame – 519

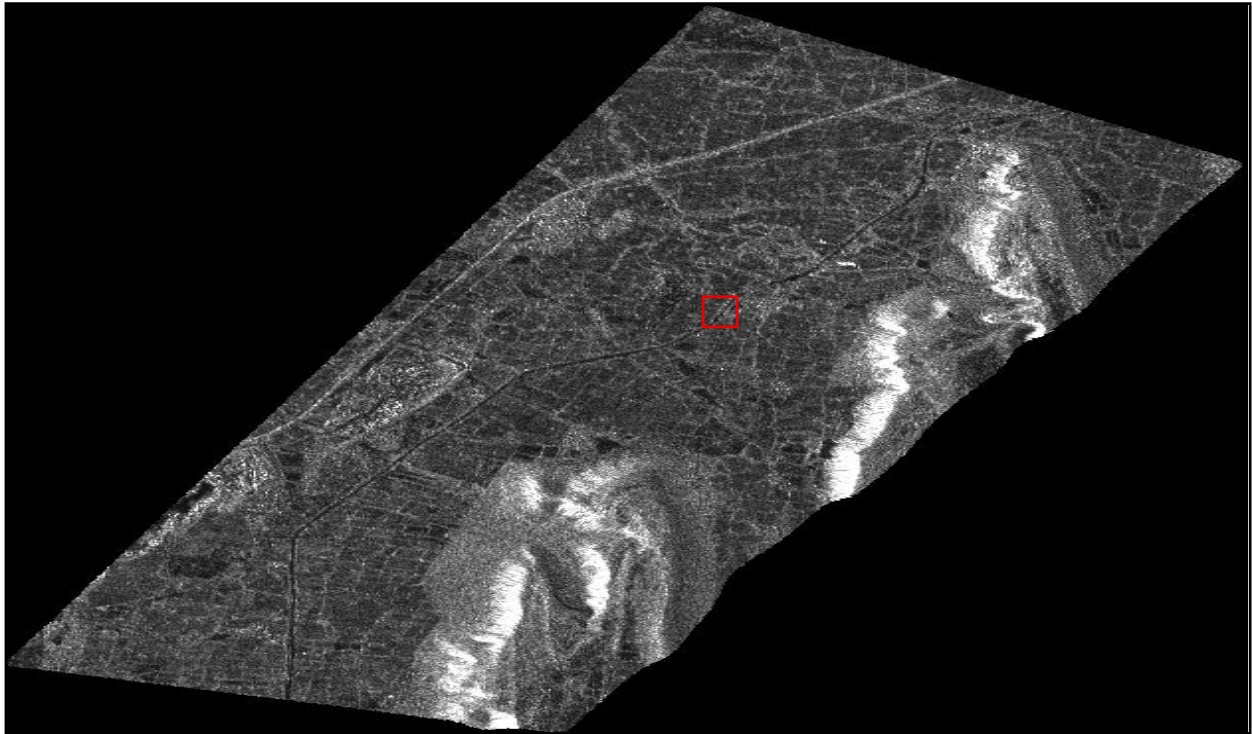
Flight Direction - DESCENDING

Polarization – VV+VH

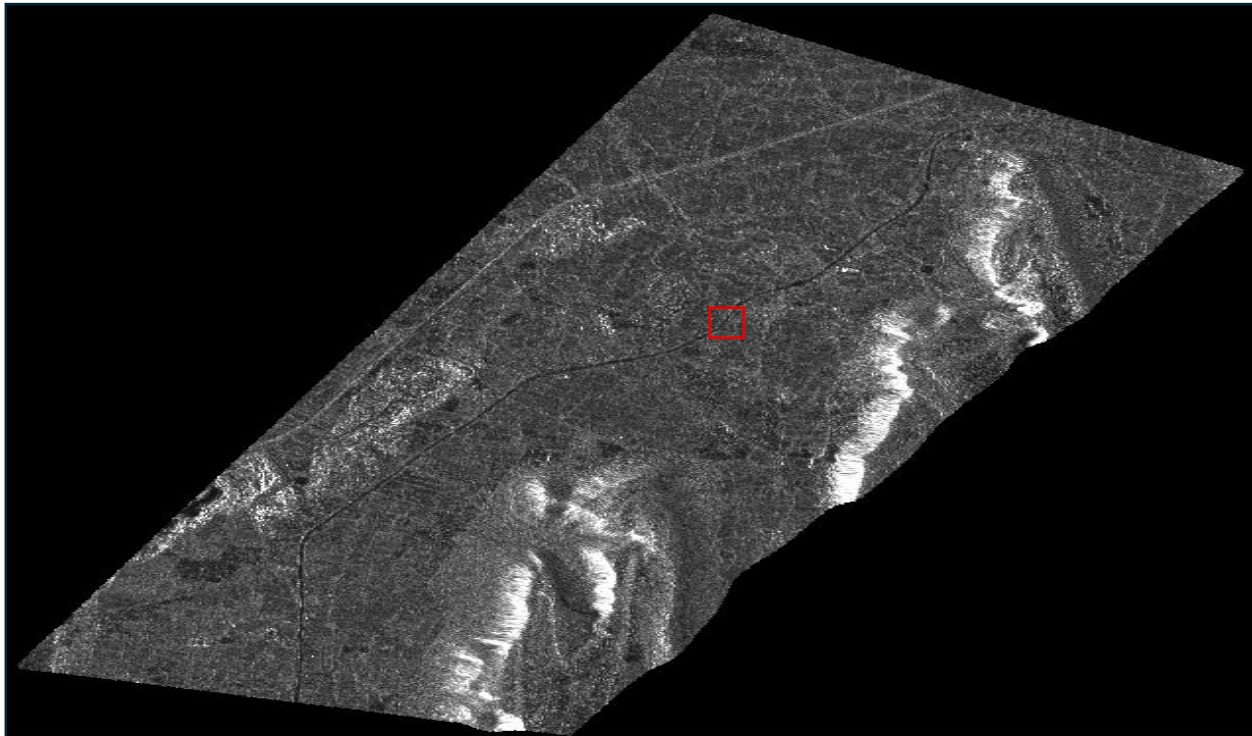
Absolute Orbit – 54706

2. Calibration, georeferencing, orthorectification of SAR images (VH/VV)

- Orthorectification of SAR image VH Polarization



- Orthorectification of SAR image VV Polarization



3. Results of SAR based bridge inspection, including visualizations

Bridge Components

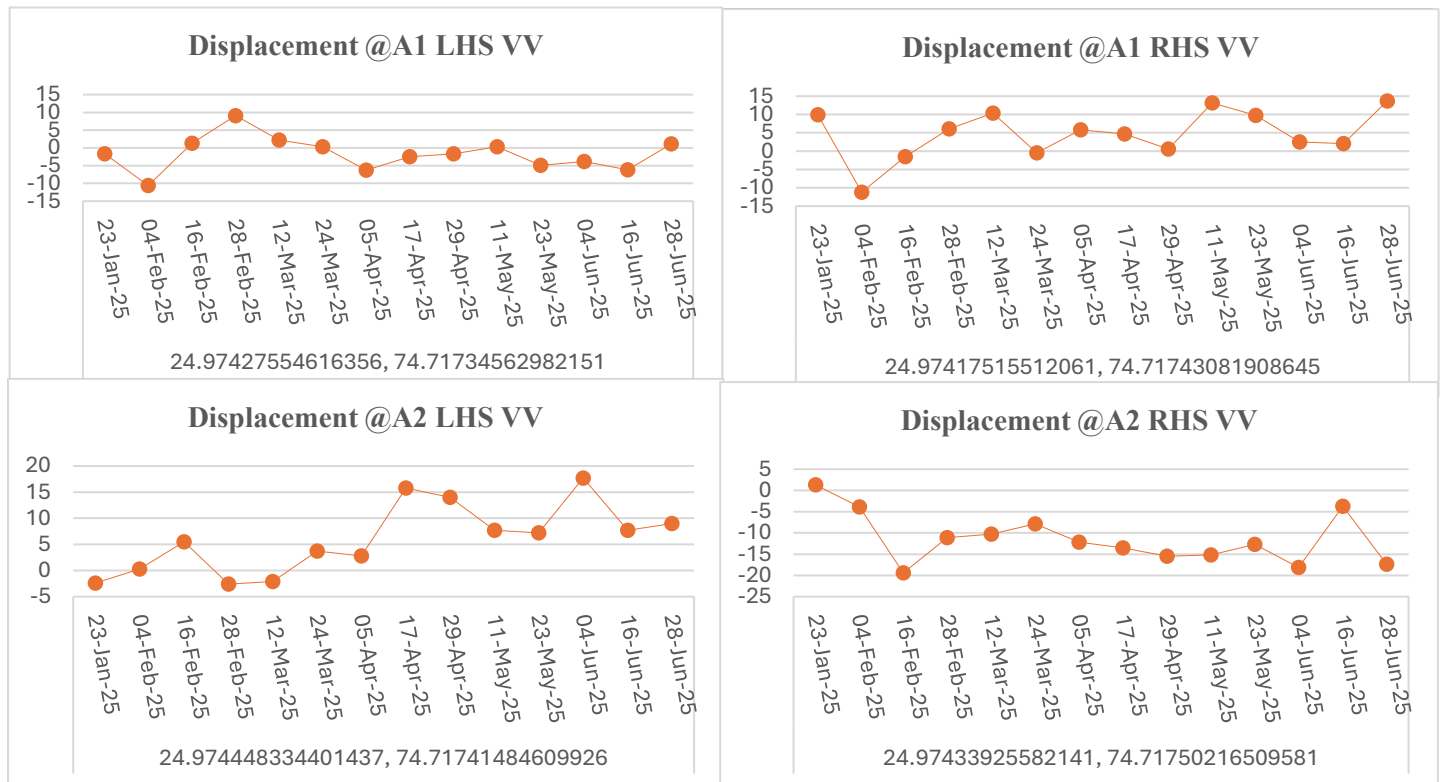
MNB – CH-914+260

Length – 20.2M

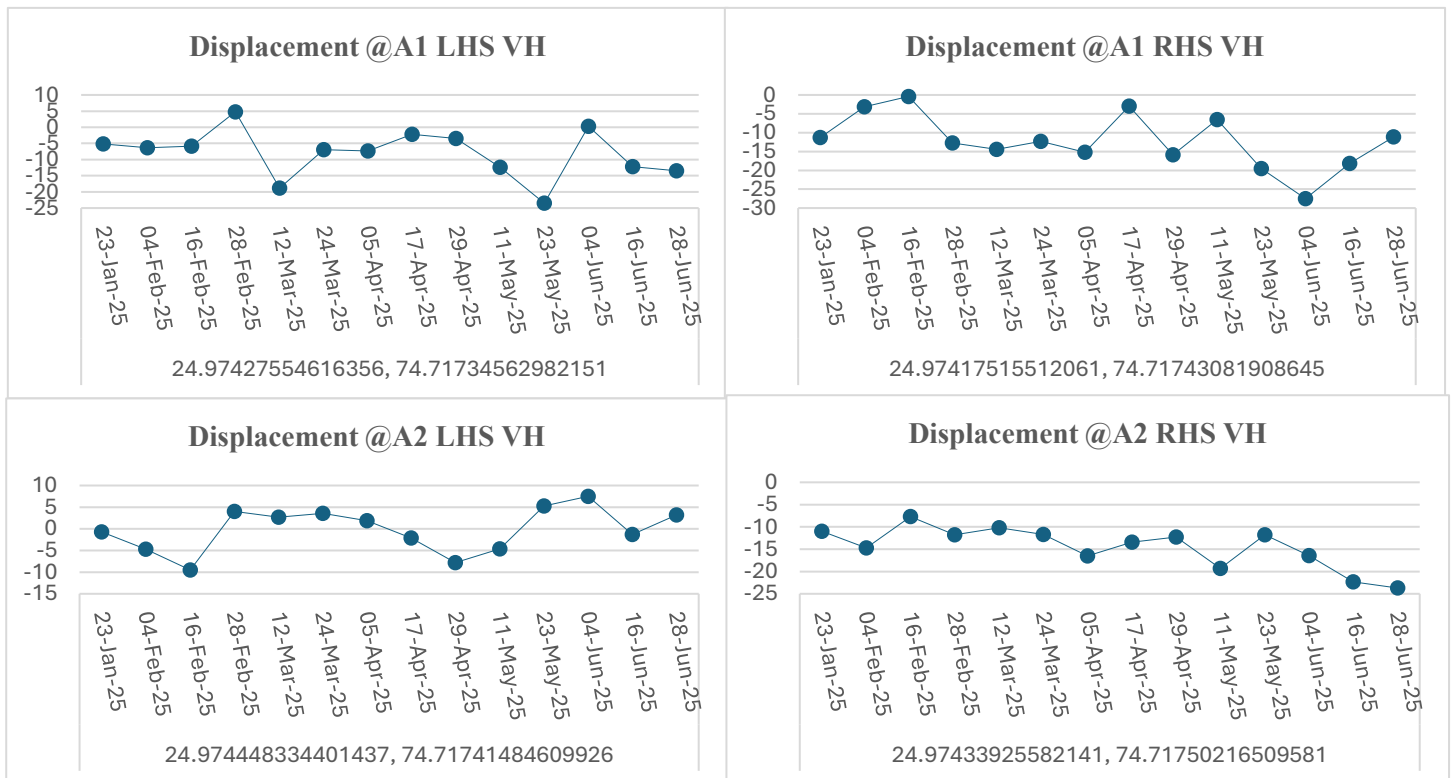


Scan Duration		From : 23 Jan 2025	To : 28 June 2025
Components & Abbreviation			
A1		Abutment 1	
A2		Abutment 2	
LHS		Left Hand Side	
RHS		Right Hand Side	

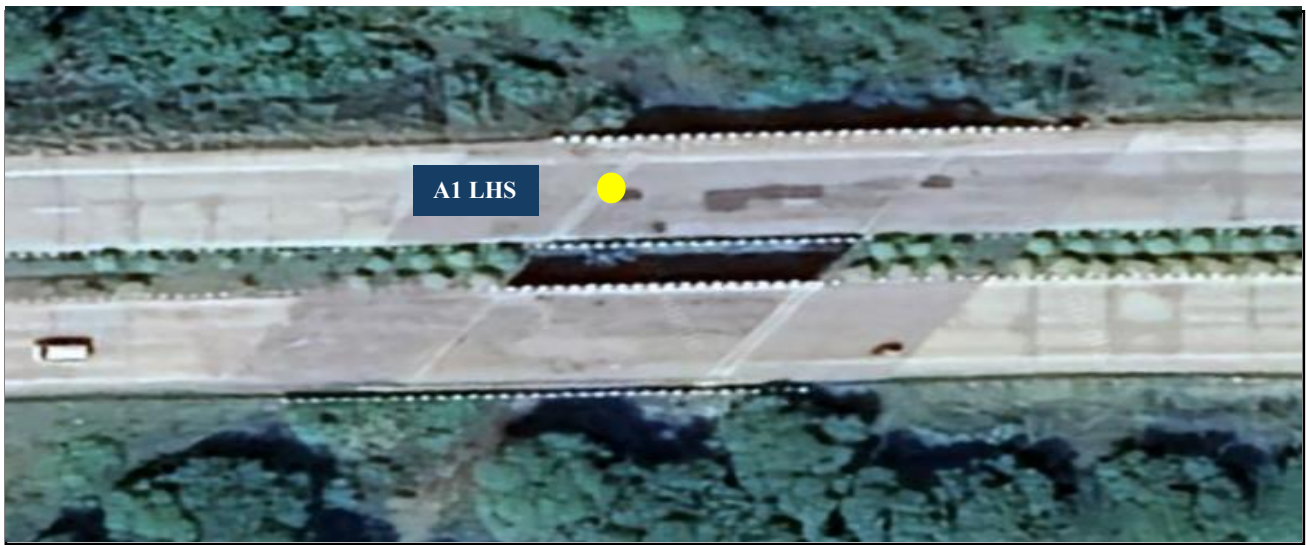
3.1 VV Displacement From 23 Jan 2025 To 28 June 2025



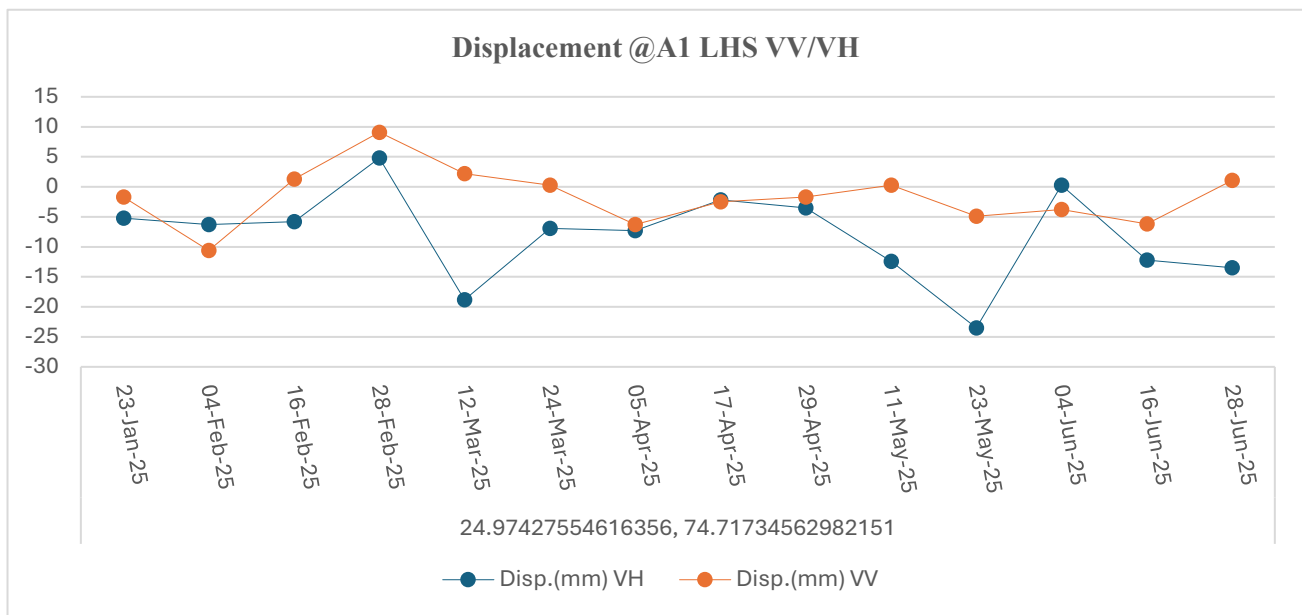
3.2 VH Displacement From 23 Jan 2025 To 28 June 2025



3.3 SAR Based Analysis at Abutment 1 LHS:



A1 LHS (24.97427554616356, 74.71734562982151)



Key Observations:

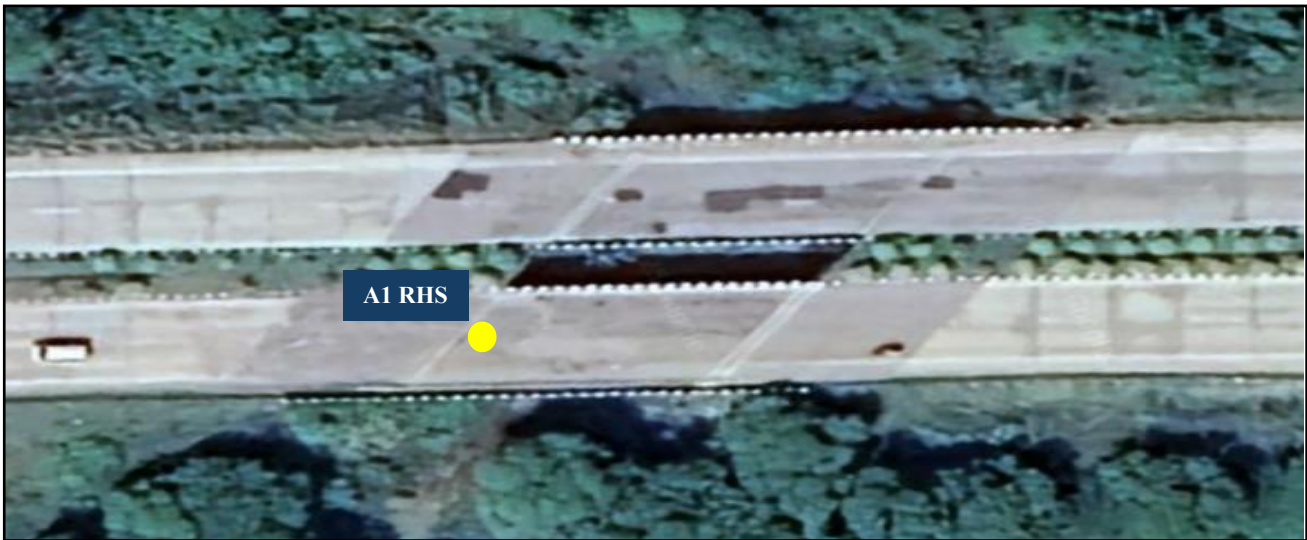
Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** Vertical displacement shows moderate variability with alternating uplift and settlement phases over the monitoring period (Jan–Jun 2025). VV values range approximately from –10 mm to +8 mm.
- **Horizontal Displacement (VH)** Horizontal displacement exhibits high variability with a strong settlement dominance, ranging roughly between +4 mm and –24 mm.

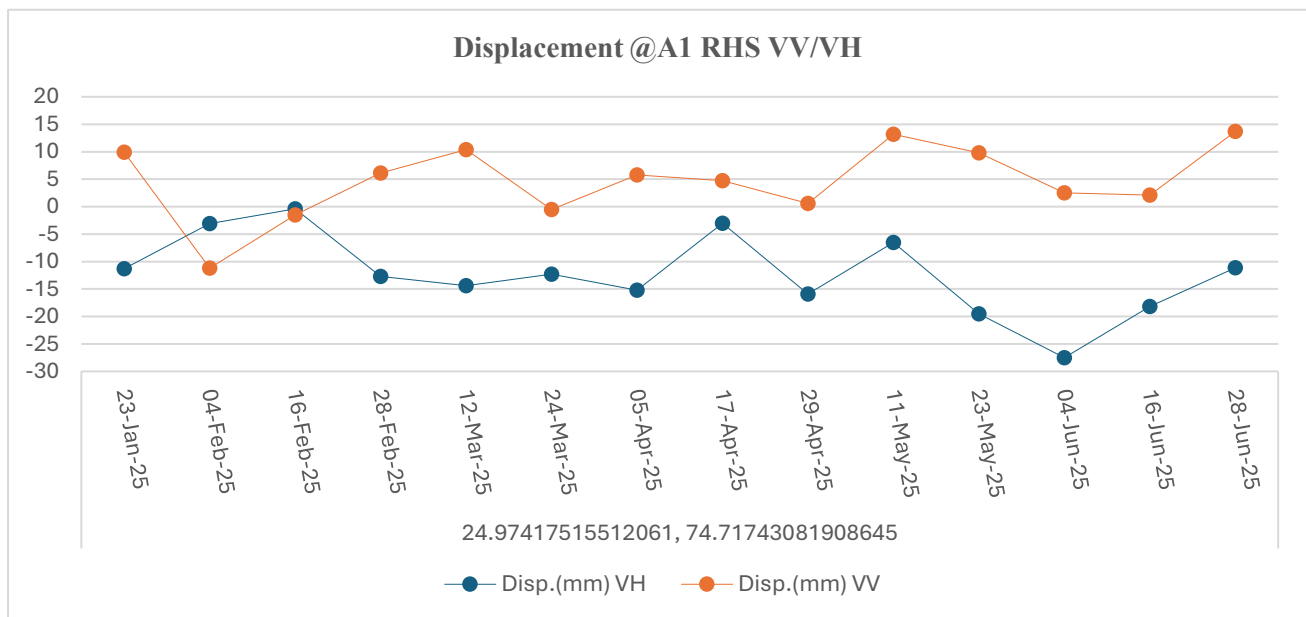
Critical Events & Anomalies:

- **January 2025:** VH starts near –5 mm, while VV is slightly negative (–2 mm), indicating early coupled lateral and vertical settlement.
- **June 2025:** VH declines again to –14 mm, while VV fluctuates between –5 mm and +1 mm, confirming continued lateral instability toward the end of the monitoring period.

3.4 SAR Based Analysis at Abutment 1 RHS:



A1 RHS (24.97417515512061, 74.71743081908645)



Key Observations:

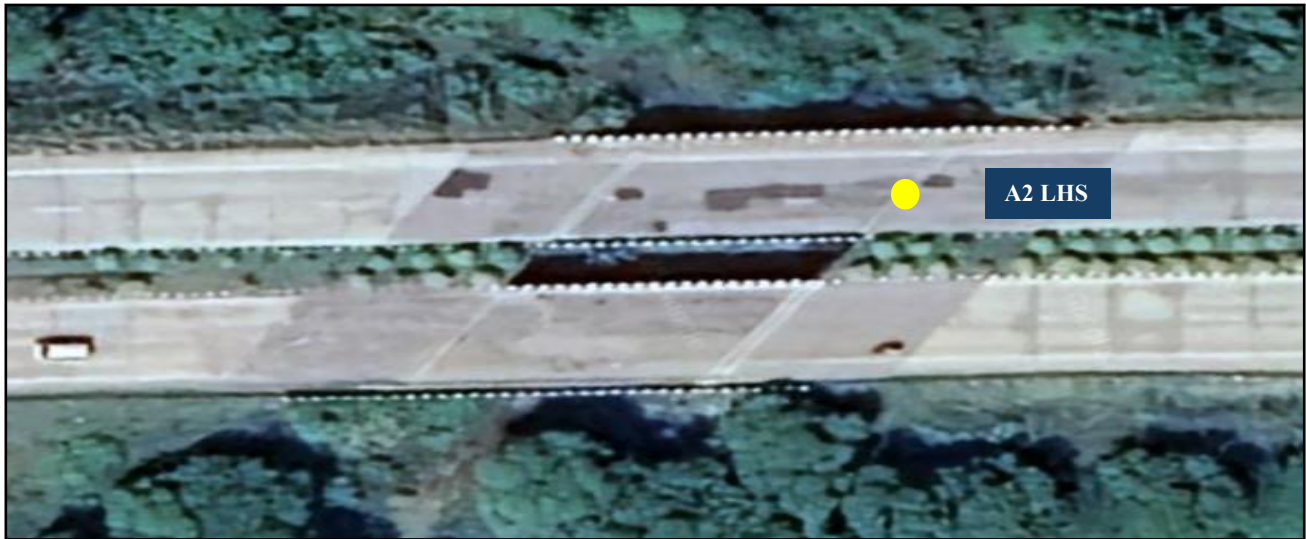
Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** Horizontal displacement shows high variability with a strong negative dominance, ranging roughly between -4 mm and -27 mm.
- **Vertical displacement (VV)** exhibits moderate variability with predominantly positive values over the monitoring period, indicating overall vertical uplift at the A1 RHS location.

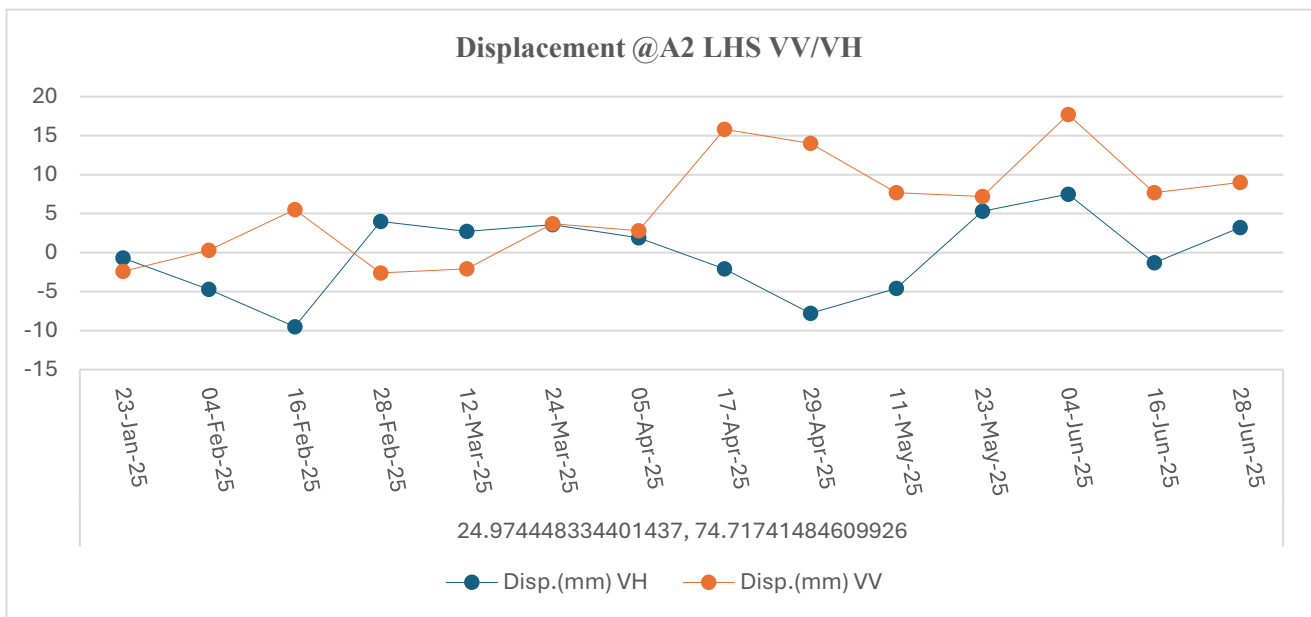
Critical Events & Anomalies:

- **January 2025:** VH starts around -11 mm, while VV is strongly positive (+10 mm), indicating early vertical uplift occurring alongside lateral settlement.
- **May 2025:** VH deteriorates further to -20 mm, while VV reduces slightly to +10 mm, indicating continued lateral instability under sustained uplift.
- **June 2025:** VH recovers marginally to -11 mm, while VV rises again to +14 mm, confirming persistent uplift with unresolved lateral settlement toward the end of the monitoring period.

3.5 SAR Based Analysis at Abutment 2 LHS:



A2 LHS (24.974448334401437, 74.71741484609926)



Key Observations:

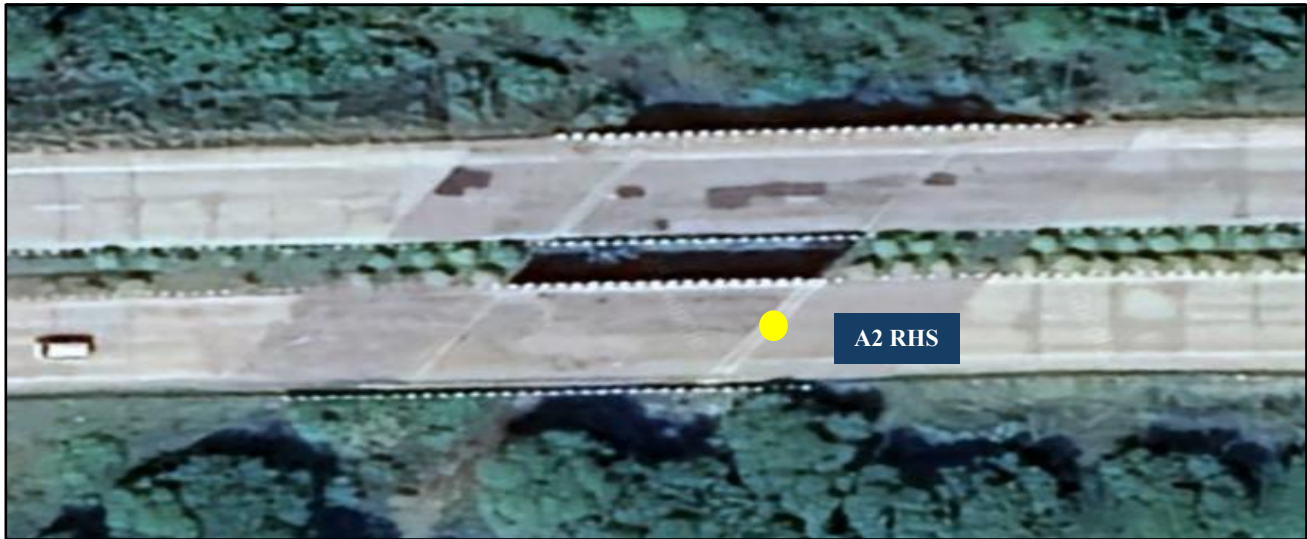
Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** Horizontal displacement shows high variability with alternating uplift and settlement phases, ranging roughly between +8 mm and -10 mm.
- **Vertical Displacement (VV)** Vertical displacement exhibits moderate to high variability with a predominantly positive trend over the monitoring period (Jan–Jun 2025), indicating overall vertical uplift at the A2 LHS location.

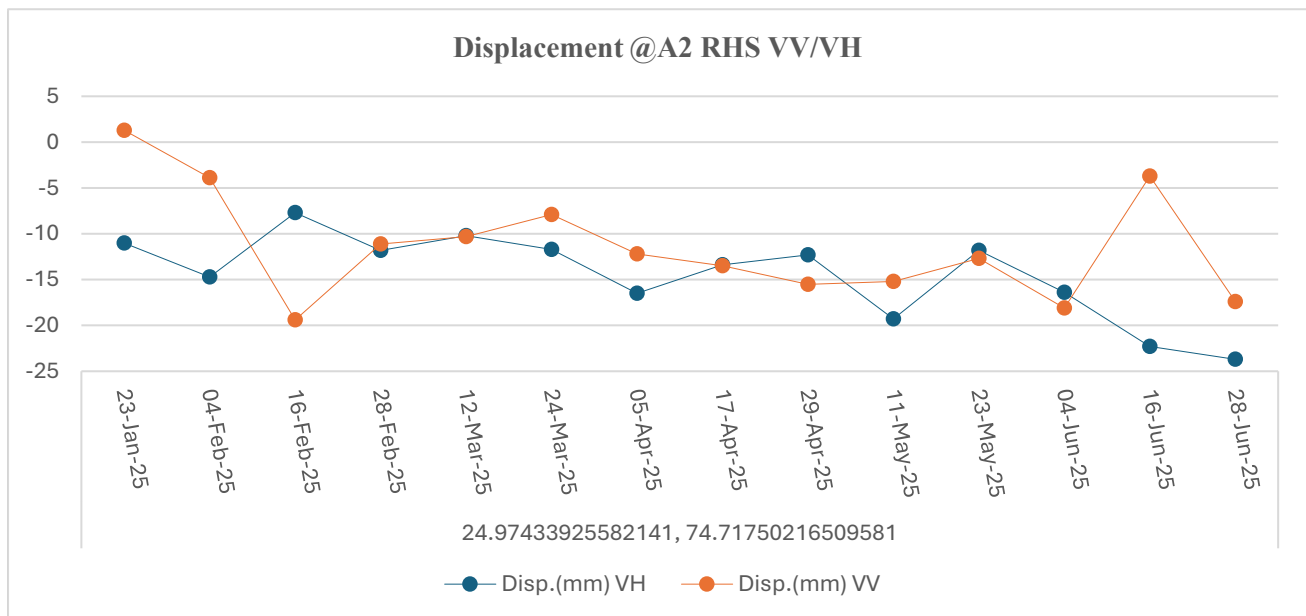
Critical Events & Anomalies:

- **January 2025:** VH starts slightly negative (-1 mm), while VV is mildly negative (-3 mm), indicating an initial low-magnitude coupled settlement phase.
- **June 2025:** VH drops back to -1 to +3 mm, while VV stabilizes around +8–9 mm, confirming continued uplift with unresolved lateral variability toward the end of the monitoring period.

3.6 SAR Based Analysis at Abutment 2 RHS:



A2 RHS (24.97433925582141, 74.71750216509581)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical (VV) displacement** Vertical displacement is predominantly negative across the monitoring period (Jan–Jun 2025), indicating overall vertical settlement at the A2 RHS location. VV values range approximately from +1 mm to –19 mm.
- **Horizontal (VH) displacement** Horizontal displacement shows moderate to high variability with a clear negative dominance, ranging roughly between –8 mm and –24 mm.

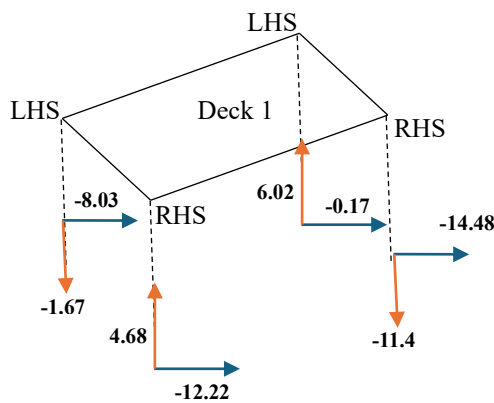
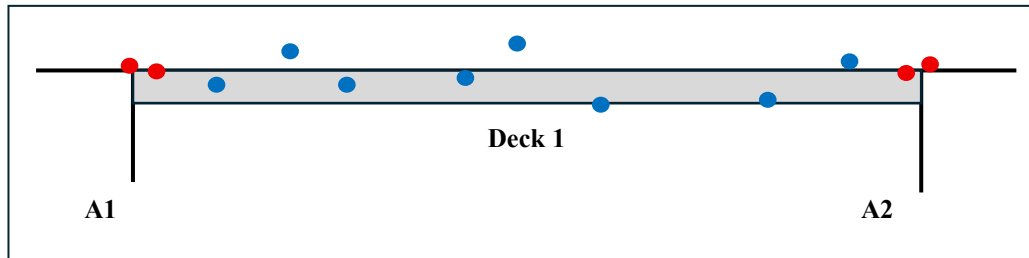
Critical Events & Anomalies:

- **April 2025:** VH reaches –17 mm, while VV declines to –13 mm, marking a distinct coupled settlement event.
- **June 2025:** VH deteriorates further to –23 to –24 mm, while VV fluctuates between –5 mm and –18 mm, confirming continued and intensifying lateral settlement with unstable vertical behaviour toward the end of the monitoring period.

ANALYSIS

1. Identify potential deformation, shifts or movement of the bridge structure

- Points of Analysis (POA) of bridge (●)



Bridge Deck Slab Analysis

Vertical Displacement Behaviour:

- The bridge deck slab exhibits non-uniform vertical displacement, indicating differential vertical response between the LHS and RHS support regions. The coexistence of uplift at the deck core and settlement at the supports confirms vertical rotation and differential bending behaviour. The maximum downward settlement is observed at the RHS lower region, reaching approximately -11.40 mm, identifying this as the most critical vertical deformation location.

Horizontal Displacement Behaviour:

- Horizontal displacement demonstrates pronounced directional variability across the deck width, reflecting torsional deformation effects. The consistent direction and magnitude differences across the deck indicate torsional rotation about a longitudinal axis. Significant horizontal displacement is also recorded at the central lower region, approximately -12.22 mm.

CONCLUSION:

- The bridge deck slab displays a torsion-influenced deformation response, characterized by moderate vertical uplift combined with significant lateral displacement. Overall, the observed displacement pattern is structurally significant and warrants continued monitoring and correlation with loading conditions and bearing performance.